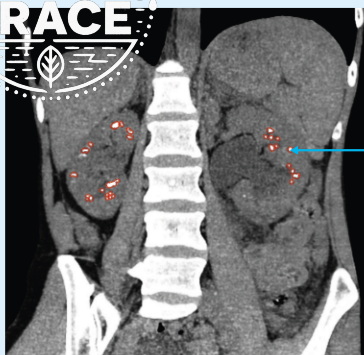
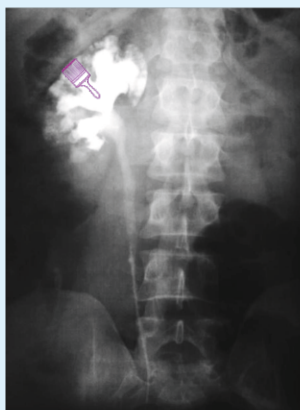




Papillary calcification in MSK

- IOC Helical CT/ NCCT abdomen: For Kidney stones and Papillary calcification



- Intravenous Pyelography (IVP)
 - No longer used d/t risk of contrast related anaphylaxis
 - **Paint brush/bouquet of flowers** appearance on IVP

PROGNOSIS

- Lifespan is not affected with Medullary sponge kidney

EXTRA MILE

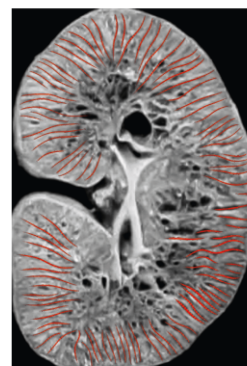
Papillary necrosis: Mnemonic - **POST CARD**

- **P** - Pyelonephritis
- **O** - Obstructive uropathy
- **S** - Sick cell anemia
- **T** - Toxins (Phenacetin → Analgesic nephropathy)
- **CA** - Chronic alcoholism
- **R** - Renal vein thrombosis
- **D** - Diabetes

ARPKD

- Defect in PKHD1 gene, Chromosome 6
- Encodes for Fibrocystin-Polyductin complex (located in CD/DCT)
- Presentation can be neonatal or later on in childhood

Radial Striations



00:13:46

CLINICAL MANIFESTATION

- Prenatal diagnosis can be made
 - 1 Echogenicity of kidneys noticed in the fetus
 - Oligohydramnios in mother
 - Cause of death: **Pulmonary hypoplasia**
- Some cases present in childhood (>10 years) with Hypertension, Uremia, Portal HTN causing variceal bleed
- Extra renal manifestations
 - **Congenital hepatic fibrosis**
 - **Caroli disease**

ADTKD

00:16:26

- Genes involved:
 - MUC-1 gene: encoding mucin-1
 - UMOD gene: encoding uromodulin
- Clinical Findings: Adult with slowly progressive kidney damage and shrunken kidneys
 1. Polyuria
 2. Anemia: Normochromic normocytic anemia
 3. Renal Function Test: Deranged, **Bland urinary sediment**
 4. ESRD: Now k/a Renal failure
 5. Hyperuricemia

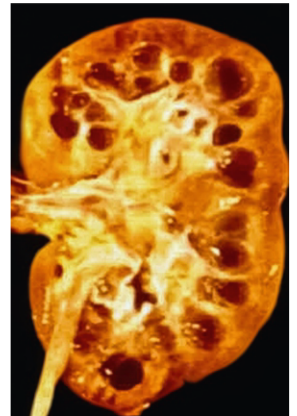
Important Information

- Earlier, Medullary Cystic Kidney Disease
 - MCKD-1: ADTKD - MUC 1
 - MCKD-2: ADTKD - UMOD

NEPHRONOPHTHISIS

00:18:55

- Autosomal recessive
- **NPHP 1 and NPHP 2** gene defect
- Tubular atrophy/ Tubular fibrosis



CLINICAL FEATURES

- Same as ADTKD
- Additional neurological features can be seen
 - Joubert syndrome
 - Agenesis of cerebellar vermis
 - MRI/CT Head: **Molar tooth sign/ Bat wing appearance**
 - Senior Loken syndrome
 - Autosomal Dominant
 - Retinitis pigmentosa





Q. ARPKD is most commonly associated with which of the following?

- a. Hepatic cysts
- b. Hepatic Fibrosis
- c. Retinal Degeneration
- d. Pancreatic cysts

Ans (b)

Q. The Hallmark feature of ADTKD on kidney biopsy is?

- a. Widespread cyst formation
- b. Interstitial fibrosis with tubular atrophy
- c. Crescentic glomerulonephritis
- d. Glomerular sclerosis

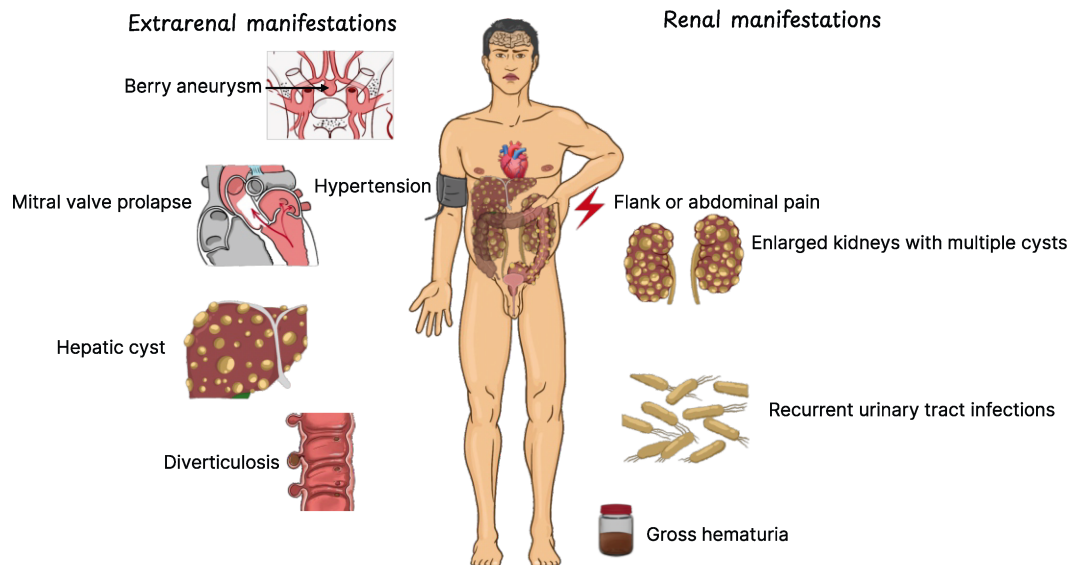
Ans (b)



ADPKD POLYCYSTIC KIDNEY DISEASE

AUTOSOMAL DOMINANT POLYCYSTIC KIDNEY DISEASE (ADPKD)

CLINICAL FEATURES



- Flank pain: **M/c presentation**
- Hepatic cyst: **M/c extrarenal manifestation**

AUTOSOMAL DOMINANT POLYCYSTIC KIDNEY DISEASE

00:02:23

- M/c life-threatening monogenic disorder
- Kidney length $\uparrow 4\times$
- Kidney weight $\uparrow 20\times$
- Peritoneal dialysis C/I: Grossly enlarged kidneys occupy the entire peritoneal space

GENETICS

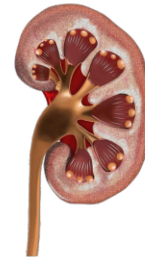
- Defective genes include
 - PKD1
 - Encoded by Polycystin 1 protein
 - Located on Chromosome 16
 - PKD2
 - Encoded by Polycystin 2 protein
 - Located on Chromosome 4
 - Other genes: GANAB, and DNA JB 11
 - Milder disease
 - Size of Kidneys can be normal/small
- Functions of Polycystin: Encode for cilia in the renal tubular cells
 - Mechanosensor- Measures the ultra- filtrate traversing through the kidneys



- Chemoreceptor- Measures the electrolytes traversing through the kidneys
- G protein signalling

AUTOSOMAL DOMINANT TUBULOINTERSTITIAL KIDNEY DISEASE (ADTKD)

- Genes involved:
 - MUC-1: ADTKD-1
 - UMOD: ADTKD-2
- Presence of cysts, located at corticomedullary junction



00:06:43

CLINICAL FEATURES OF ADPKD

Note- Complete penetrance with variable expression

1. Asymptomatic → HTN (↑Renin) later on
2. Flank pain
 - M/c symptom
 - D/t bleeding/infection in the cyst → ↑ pressure in the cyst
3. Gross hematuria: Rupture of cyst connected to the collecting system
4. Urosepsis risk: **Pyelonephritis**
 - Examination finding → Costovertebral tenderness
 - Urine examination- WBC cast, $>10^4$ CFU/ML
5. Nephrolithiasis: Uric acid crystals (higher incidence)
6. RCC risk is same as in normal population but if it occurs, it can be multicentric and B/L

00:08:25

Important Information

- Leading Causes of Death in ADPKD: HTN related complications (Cardiovascular Mortality)

7. Extrarenal Manifestations

- Hepatic cyst, Splenic cyst and Pancreatic cyst
→ M/c extrarenal manifestation
- CNS: Intracranial aneurysm, Arterial dolichoectasia > Arachnoid cysts
- CVS: Mitral valve prolapse
- GIT: Colonic diverticulosis (causes constipation), abdominal wall hernia

Important Information

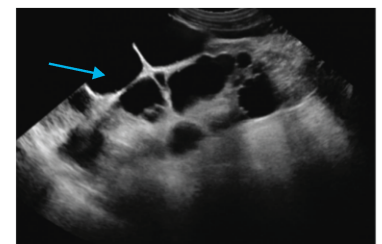
- M/c extrarenal manifestation in ADPKD: **Hepatic Fibrosis**

WORK UP

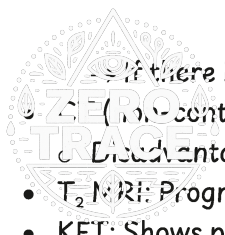
00:14:08

- Ultrasound abdomen: **Preferred imaging modality** for both initial screening and diagnosis
 - **Pei criteria**
→ If there is a positive family history of ADPKD

AGE (YEARS)	MINIMUM CYST REQUIRED (TOTAL IN BOTH KIDNEYS)	SENSITIVITY %
16-40	≥ 3 cysts (Unilateral or Bilateral)	100%
40-59	≥ 2 cysts per kidney	100%
≥60	≥ 4 cysts per kidney	100%



Multiple Cysts



- If there is no family history of ADPKD: ≥ 10 cysts in each kidney on MRI/CT is required for diagnosis
- **CT (No contrast):** Nephrolithiasis
- **Disadvantage:** Radiation
- **T₂ MRI:** Progression of disease
- **KFT:** Shows progressive derangement, $\downarrow$ eGFR
- **Next-generation sequencing:** Mutation analysis
 - **Overall best investigation for diagnosis**
 - Identifies the defective gene (PKD-1/ PKD-2)

TREATMENT

00:18:54

- HTN control: ACE inhibitors
- Lipid-soluble antibiotics: Trimethoprim-Sulfamethoxazole
- Analgesics/narcotic analgesics/TEN (Transcutaneous electrical nerve stimulation)
- Pre-transplant nephrectomy
 - For pain relief before allogenic kidney transplantation
- Vaptans: Slows growth of cysts which is shown to be mediated by Vasopressin
- Sirolimus: Inhibits the upregulated mTOR pathway that causes epithelial cell proliferation
- Octreotide

Important Information

- **Lanreotide** (long acting Octreotide) injections are used in management of Acromegaly

AUTOSOMAL RECESSIVE POLYCYSTIC KIDNEY DISEASE (ARPKD)

00:21:31

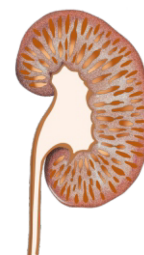
- Physical appearance: **Radial striations** in kidney

CLINICAL FEATURES

- Hypertension
- End stage renal disease
- Uremic manifestation

EXTRARENAL FEATURES

- Hepatic fibrosis
- Caroli disease



BIRT HOGG DUBE SYNDROME

00:22:44

FEATURES	DESCRIPTION
Genetics	• AD • Caused by FLCN gene mutation (Tumor suppressor gene)
Lung Involvement	• Multiple lung cysts, especially at the lung base $\rightarrow$ Spontaneous pneumothorax
Skin findings	• Fibrofolliculomas: Benign hair follicle tumors on face, neck, upper torso • Trichodiscomas • Acrochordons (skin tags)
Renal Tumors	• 15-30% risk of renal cell carcinoma (RCC)- Mostly chromophobe or hybrid oncocytomas
Cancer Risk	• Increased risk of Renal cancer, but not other cancers



- Q. Which of the following statements regarding extrarenal manifestations of autosomal dominant polycystic kidney disease (ADPKD) is correct?
- a. Lung cysts are commonly seen in ADPKD patients
 - b. Brain parenchyma is frequently affected by cyst formation in ADPKD
 - c. Arachnoid cysts are not associated with ADPKD
 - d. Mitral valve prolapse with mid-systolic clicks

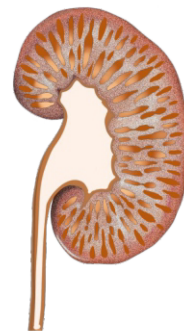
Ans. (d)

- Q. Which of the following is not an extra-renal manifestation of ADPKD?
- a. Tricuspid regurgitation
 - b. Berry aneurysm
 - c. Diverticulosis
 - d. Caroli disease

Ans (d)

- Q. Which of the following is a renal disorder that shows the following appearance of the kidney?
- a. Autosomal dominant polycystic disease
 - b. Autosomal recessive polycystic disease
 - c. Medullary sponge kidney
 - d. Autosomal dominant tubulointerstitial kidney disease

Ans (b)





THROMBOTIC THROMBOCYTOPENIC PURPURA



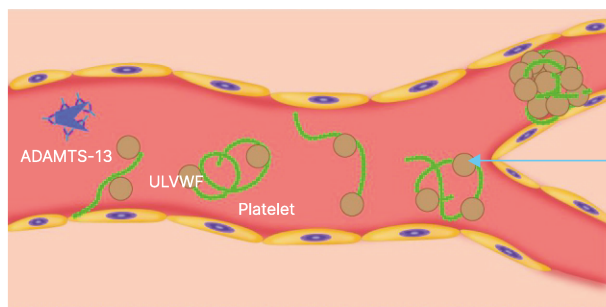
THROMBOTIC THROMBOCYTOPENIC PURPURA

INTRODUCTION

00:00:45

Normal function

- ADAMTS13 cleaves large vWF multimers into smaller fragments, preventing excessive clot formation



ADAMTS-13 cleaving ULVWF into smaller VWF multimers

TTP

↓ ADAMTS13 activity (due to autoantibodies)



Uncleaved large vWF multimers



Spontaneous platelet aggregation

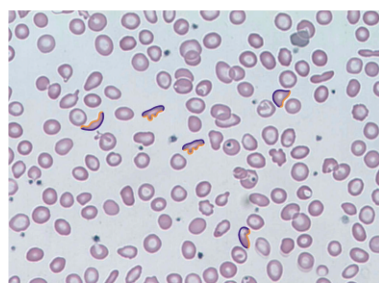
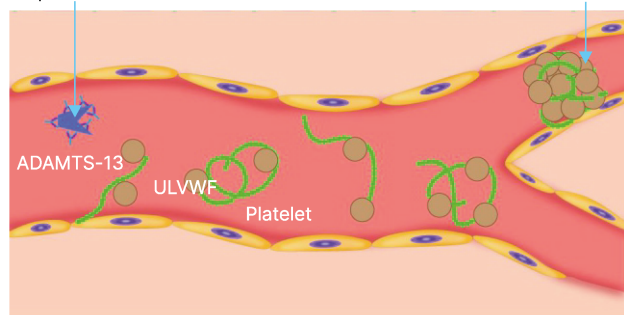


Formation of multiple platelet-rich microthrombi

- Results in:
 - Thrombocytopenia
 - Schistocytes/Helmet cells on peripheral smear

Presence of autoantibodies against ADAMTS-13 causing impairment/Reduced ADAMTS-13

Platelet aggregation on ULVWF forming occlusive platelet thrombus



Schistocytes/Helmet Cells

PATHOGENESIS

- ADAMTS13:
 - Full form: A Disintegrin and Metalloproteinase with Thrombospondin Motif 13
 - Function: Break down of ultra- large multimers of vWF

DEFINITION

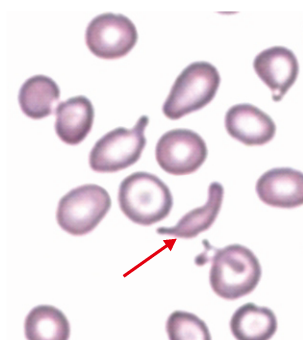
- Microangiopathic Hemolytic anemia (MAHA) + Low ADAMTS13 activity (5-10% of normal)



- **A** - Anemia
 - Microangiopathic hemolytic anemia
 - Coombs negative Hemolytic anemia
 - ↓Hb, ↑LDH, ↑Unconjugated Bilirubin
- **T** - Thrombocytopenia
- **N** - Neurological features
 - Confusion, drowsiness, vision loss, ataxia, and seizures
 - Manifestations of **small vessel stroke**
- **RI** - Renal insufficiency
 - Elevated serum creatinine

WORKUP FOR TTP

- NCCT Head
 - Initially normal
 - After 6 hours, multiple hypodensities
- Coagulogram (platelets are consumed)
 - Bleeding Time (BT): ↑
 - PT: Normal
 - aPTT: Normal
 - Fibrinogen: Normal
- CBC: ↓ Hb
- LDH: ↑↑
- LFT: Unconjugated bilirubin ↑
- Peripheral smear: Schistocytes

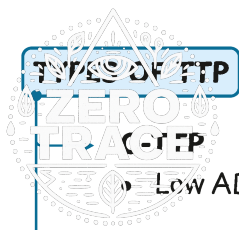


00:05:59

TREATMENT OF TTP

- **iTTP (autoimmune)**
 - Plasma exchange
 - Prednisolone
 - Rituximab
 - **Caplacizumab**: MOA → Inhibits vWF binding to the platelets (inhibit formation of platelet microthrombi)
- **Drug-Induced TMA**
 - Discontinue offending drugs
- **c-TTP: Congenital deficiency of ADAMTS13**
 - Plasma transfusion

00:07:45



- Low ADAMTS13 since birth called as **Upshaw-Schulman syndrome**

→ **iTTP**

- Immunological
- Increased clearance of ADAMTS13

→ **Drug-induced Thrombotic Microangiopathy**

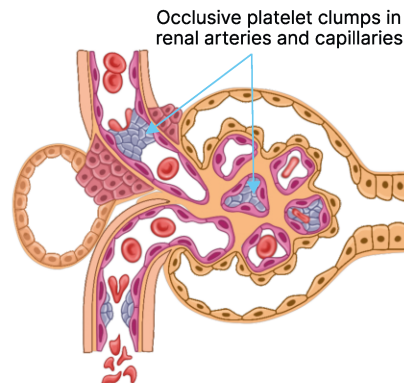
- Mitomycin C
- Bortezomib, Interferon
- Tacrolimus, Sirolimus
- **Clopidogrel**
- Quinine, Bevacizumab

→ **Infections, Pancreatitis, Malignancy, Pregnancy**

- Neutrophils inhibit activity of ADAMTS13

HEMOLYTIC UREMIC SYNDROME (HUS)

- MAHA + Renal insufficiency
- Diarrhea/Dysentery: **Shiga toxin-producing E.coli** (O157:H7)
- Neurological symptoms:
 - Encephalopathy, Seizures, Dysphasia and Perseveration (repetition of the same statement)
- Treatment:
 - Stop antibiotics
 - Packed RBCs
 - Hemodialysis
- Peripheral smear findings- Schistocytes



Schistocytes-Red Cell damage by occlusive platelet clumps and MAHA

ATYPICAL HUS (aHUS)

- Complement-mediated HUS
- Complement factor H deficiency
- Treatment: **Eculizumab**

TTP vs. D⁺ HUS

D ⁺ HUS	TTP
• Child with diarrhea • MAHA • AKI (Severe) • No Fever	• Adult iTTP, Pregnancy, Pancreatitis • MAHA • AKI (Mild) • Fever

- Platelets ↓
- Shiga toxin in stool
- Encephalopathy, Perseveration, Seizures, Memory defect
- Treatment: Supportive care, Hemodialysis

- Platelets ↓
- Low ADAMTS13 activity
- Small vessel stroke with aphasia, vision loss, ataxia, seizure
- Treatment: Plasma Exchange

COMPARISON TABLE OF ITP, HSP, TTP, DIC

00:15:15

FEATURE	ITP	HSP	TTP	DIC
Cause	Autoimmune	IgA vasculitis	ADAMTS13 deficiency/ ↑ Clearance	Sepsis/shock
BT	↑	Normal	↑	↑
PT	Normal	Normal	Normal	↑
aPTT	Normal	Normal	Normal	↑
Fibrinogen	Normal	Normal	Normal	↓

- HSP- A child presenting with purpura on the buttocks or thigh (extensor purpura) and later on spreads anteriorly

EXTRA MILE



Non-palpable purpura



Palpable Purpura-HSP

- Non-palpable purpura: **ITP**
- Palpable purpura: **HSP**



Q. Which of the following is not a feature of TTP?

- a. Low platelets
- b. Fever
- c. Increased LDH
- d. Large vessel thrombosis

Ans (d)

Q. A 5-year-old boy presented with bloody diarrhea since the past few days. He was admitted to the inpatient ward due to low urine output. Testing showed anemia, thrombocytopenia and helmet cells on peripheral smear and increased levels of BUN. Which of the following is not used for management of this case?

- a. Hemodialysis
- b. Plasmapheresis
- c. PRBC
- d. Furosemide

Ans (b)



KIDNEY URINE ANALYSIS



PROTEINURIA

00:00:34

- Normal amount of protein excreted in urine: <150 mg/day
 - Through Glomerulus: **Albumin** (<30 mg/day)
 - Through Tubules: **Uromodulin / Tamm Horsfall protein** (120 mg/day)
- Estimation of albumin:
 - Estimated using Spot urine sample:

$$\text{Albumin excretion rate} = \frac{\text{Urine albumin}}{\text{Urine Creatinine}} \text{ mg/dl}$$

	MODERATE ALBUMINURIA	SEVERE ALBUMINURIA
Albumin excretion rate (AER)	• 30-300 mg albumin per gm of urinary creatinine	• >300 mg albumin per gm of urinary creatinine
Causes	• Diabetic nephropathy • Hypertension (malignant) • Glomerulonephritis	• Diabetes mellitus • HTN • Glomerulonephritis • Multiple myeloma/ Paraproteinemia, MGUS (Monoclonal gammopathy of undetermined significance) • Fever, Exercise, Orthostatic Transient rise

URINE ALBUMIN-TO-CREATININE RATIO (UACR) CATEGORIES (in mg/g and mg/mmol)

CATEGORY	UACR (mg/g)	UACR (mg/mmol)
Normal (A1)	<30 mg/g	<3 mg/mmol
Moderately increased (A2) (Previously microalbuminuria)	30-300 mg/g	3-30 mg/mmol
Severely increased (A3) (Previously macroalbuminuria)	>300 mg/g	>30 mg/mmol

	OVERLOAD PROTEINURIA	FUNCTIONAL PROTEINURIA
CAUSE	• Multiple Myeloma • Rhabdomyolysis • Hemoglobinuria	• Jogging/ Exercise



NEPHROTIC RANGE PROTEINURIA

00:09:15

• **UPEP**: 24 hr: 24 hr timed specimen

• **> 2 gm/g** of urinary creatine: Spot sample

CAUSES

- Diabetic nephropathy
- Amyloidosis
- Focal segmental glomerulosclerosis (FSGS)
- Membranous glomerulopathy
- Minimal change disease/Lipoid nephrosis

URINE SAMPLE

- Nephrotic range proteinuria will have **foamy urine**
- Urine turns yellow on-air exposure → d/t **urochromes**
- **UPEP (Urinary Protein Electrophoresis)** identifies the type of protein (albumin vs globulin)



5 minutes



15 minutes

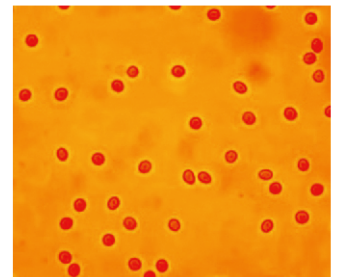
URINE DIPSTICK TEST

	PROTEIN LEVELS
Trace	15-30 mg/dl
1+	30-100 mg/dl
2+	100-300 mg/dl
3+	300-1000 mg/dl
4+	>1000 mg/dl/1gm

HEMATURIA

00:15:05

- Normal amount: **<3 RBC/HPF** in centrifuged sediment of urine sample
- Definition:
 - **>5 RBCs/HPF** in a centrifuged urine sample, confirmed in at least **3 separate reports** at weekly intervals
 - Or
 - **>100 RBCs/HPF** in a single report
- **Dysmorphic RBCs** in urine microscopy indicate glomerular bleeding



RBCs seen on urine microscopy

CAUSES

- Leading cause of (gross) hematuria: **Kidney stones**
- Leading cause of Glomerulonephritis leading to microscopic hematuria: **IgA nephropathy / Berger's disease**

CAUSE OF RED URINE	CAUSE OF COLA COLOR URINE WITH RBC CASTS AND DYSMORPHIC RBC
• Beetroot ingestion	• Glomerulonephritis

- Paraphuria
- Hematuria
- Myoglobinuria (Rhabdomyolysis)
 - Creatinine kinase values

- Good pasture: Hematuria + Hemoptysis
- Alport syndrome
 - Defective type 4 collagen
 - Features: Anterior lenticonus, SNHL, End stage renal disease
- Lupus nephritis
 - Light microscopy finding of wire loop lesions

Dipstick test positive for RBCs

CENTRIFUGE SPECIMEN @ 1500RPM FOR 5 MIN

Supernatant Clear

Hematuria

Supernatant Red

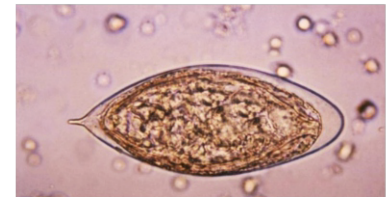
After 3 hrs
Clear

Myoglobinuria

After 3 hrs
Red

Hemoglobinuria

- Hematuria + Pyuria/Bacteriuria: **UTI**
- Hematuria + RBC cast/Dysmorphic RBC + Proteinuria: **Glomerulonephritis**
- Katayama fever: Hematuria + fever
 - Cause: Schistosomiasis
 - Urine M/E: Egg with **terminal spine** → **Schistosoma haematobium**



HEMATURIA VS MYOGLOBINURIA VS HEMOGLOBINURIA

00:25:35

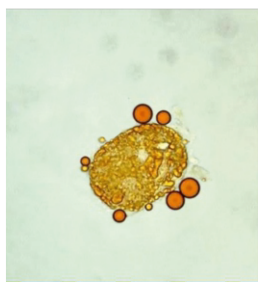
FEATURE	HEMATURIA	MYOGLOBINURIA	HEMOGLOBINURIA
URINE COLOUR	Cola/Tea	Cola/Brown	Red/Pink
URINE MICROSCOPY	RBCs, RBC Cast	No RBCs	No RBCs
SERUM CK	Normal	11 (>1000 U/L)	Normal

EOSINOPHILURIA

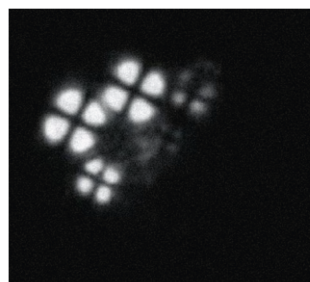
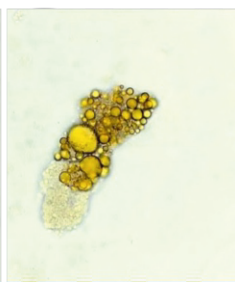
00:26:18

CAUSES

- Atheroembolic kidney disease (**Post coronary angiography**)
 - AKI d/t dissemination of atherosclerotic plaque fragments lodging in to Renal arteries during angiography
 - Features
 - ↑↑KFT post coronary angiography
 - Toe pain
 - Purpura
 - Urine M/E: Eosinophiluria
- Allergic interstitial nephritis
 - AKI **post IV Vancomycin** administration
 - Triad: Fever + rash + eosinophiluria (**Hansel Stain**) in patient who develops AKI after antibiotic (Vancomycin), PPI, Allopurinol, SLE



Oval fat bodies in urine



Maltese cross appearance in urine

- **Oval fat bodies** in urine seen in
 - **Nephrotic syndrome**
 - Fat embolism syndrome
 - Chyluria
- Maltese cross appearance in urine: **Nephrotic syndrome**
- Maltese cross appearance in Peripheral smear: **Babesia Microti** (vector is **Ixodes**)
- Maltese cross appearance in CSF: **Cryptococcal neoformans meningitis**

SPECIFIC GRAVITY OF URINE

00:33:12

- Normal specific gravity: 1.005 - 1.030
- Urine osmolality: 100 - 900 mOsm

ISOSTHENURIA

- Fixed urine specific gravity: 1.010 with inability to concentrate or dilute the urine
- Causes: **SIC**
 - **S** - **Sickle cell anemia**
 - **I** - Ischemic or toxin induced ATN (late stage)
 - **C** - CKD/ Chronic glomerulonephritis

pH OF URINE

00:34:36

- Normal pH of urine: 4.5 - 8.0
- If pH > 5.5 in setting of hypokalemic NAGMA, indicates **RTA-1**
 - Inability to acidify the urine d/t damage to alpha intercalated cells
- If urine pH < 5.5 in setting of hypokalemic NAGMA and increased fractional excretion of bicarbonate, indicates **RTA-2**

COLOR OF URINE

00:36:10



Pink Urine



Red Urine



Black Urine

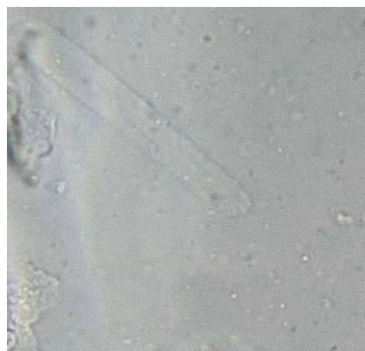


PINK URINE	RED URINE	BLACK URINE
• Debaroxamine • <i>Serratia marcescens</i> (Pink diaper syndrome)	• Beetroot ingestion • Porphyrria	• Black urine becomes less black as day progresses → Paroxysmal nocturnal hemoglobinuria • Urine turns black on air exposure → Alkaptonuria • Cola colored urine → hematuria

CASTS

00:37:43

HYALINE CASTS	• Normal finding • Composed of Tamm Horsfall Protein / Uromodulin
RBC CAST	• Acute glomerulonephritis
WBC CAST	• Pyelonephritis • Acute interstitial nephritis
MUDDY BROWN CAST	• Acute tubular necrosis
BROAD CAST	• CKD (caused by Diabetic nephropathy)
GRANULAR WAXY CAST	• Chronic glomerulonephritis



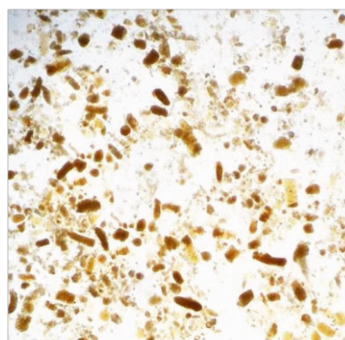
Hyaline cast



RBC cast



WBC cast



Muddy Brown cast



Broad cast



Waxy cast

CRYSTALS IN URINE MICTOSCOPY

00:39:44

ENVELOPE SHAPED	• Envelope shaped crystals • Crystal: Calcium Oxalate dihydrate • Can be seen in normal people due to eating spinach or Kidney stones
DUMB BELL SHAPED	• Calcium Oxalate monohydrate crystals • Seen with Ethylene glycol poisoning
COFFIN LID APPEARANCE	• Struvite stone • Reason: Urinary alkalization by <i>Proteus mirabilis</i>
HEXAGONAL	• Seen with Cystinuria
ROSETTE SHAPED	• Calcium phosphate



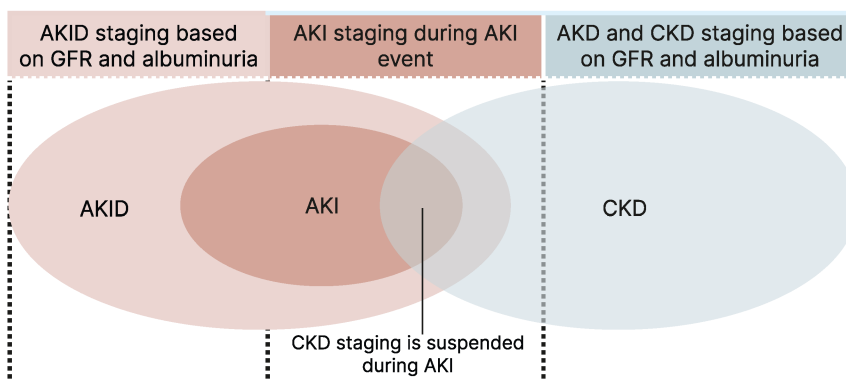
Q. Diabetic patient on vancomycin develops fever, rash and anuria. Urine routine microscopy shows eosinophiluria, with elevated BUN and serum creatinine 4 mg/dl. which is most likely cause?

- a. Diabetic nephropathy
- b. Allergic interstitial nephritis
- c. Atheroembolic kidney disease
- d. Rapidly progressive Glomerulonephritis

Ans: (b)



33. ACUTE KIDNEY INJURY



- CKD: eGFR <60 ml/min/1.73 m² BSA for >3 months
- AKD: eGFR <60 ml/min/1.73 m² BSA for <3 months
- AKI : ↑ S. Creatinine x <7 days

INTRODUCTION

00:03:48

- Acute kidney injury (AKI) complicates **30%** of all ICU admissions

COMMUNITY-ACQUIRED AKI	HOSPITAL-ACQUIRED AKI (CUT-OFF: >48 HRS)
• Tropical infections: Diarrhea, Cholera • Environmental injury: ◦ Heat stroke ◦ Burns • Obstetric complications • Envenomation • Drugs from alternative systems: ◦ Aristolochic acid: Used in herbal medicines	• Sepsis • ARDS → MODS (≥1 organ failure) • Prescription drugs: ACE inhibitors/ ARBs, NSAIDs, Amphotericin-B • Diagnostic procedures: Contrast induced nephropathy

STAGES OF AKI

00:07:57

STAGE	SERUM CREATININE	URINE OUTPUT (1ML/KG/HR)	DURATION
1	• Rise over baseline ◦ >1.5-1.9 times above the normal value within 7 days ◦ ≥0.3 mg/dL rise within 48 hours	<0.5 mL/kg/hr	>6 hrs
2	• 2.0-2.9 times ↑	<0.5 mL/kg/hr	>12 hrs
3	• >3.0 times of normal or S. Creatine >4.0 mg/dL or RRT initiated	<0.3 mL/kg/hr	≥24 hrs